MANUFACTURING ANALYTICS CAPSTONE PROJECT

AI-Powered Predictive Maintenance System for Manufacturing Equipment

Predicting milling-machine failures from operating data using machine learning, explainable AI and a maintenance decision framework

Submitted by

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Dataset

AI4I 2020 Predictive Maintenance Dataset (Matzka, 2020) — 10,000 records, 14 variables

Tools

Python · Pandas · NumPy · Matplotlib · Seaborn · Scikit-learn · SHAP · Power BI

Abstract

Unplanned equipment failure is one of the most expensive problems in manufacturing: it stops production, delays deliveries and pushes maintenance cost up. This project builds a machine-learning system that predicts whether a milling machine will fail, using six readings that are available before the failure occurs — product type, air temperature, process temperature, rotational speed, torque and tool wear.

The work uses the public AI4I 2020 Predictive Maintenance dataset (10,000 records, 339 failures, i.e. 3.39%). After confirming the data is complete and free of duplicates, exploratory analysis showed that failed machines run at higher torque (about 50.2 Nm against 39.6 Nm) and higher tool wear (about 144 min against 107 min), while temperature differences are small. Six classifiers — Logistic Regression, K-Nearest Neighbours, Decision Tree, Random Forest, Support Vector Machine and Gradient Boosting — were trained on a stratified 80/20 split and tuned with cross-validated grid search. Because only 3.4% of records are failures, models were judged on failure-class precision, recall and F1-score rather than accuracy.

The **tuned Decision Tree** was selected as the final model. On the 2,000-record test set it reached **98.55% accuracy, 79.1% precision, 77.9% recall, an F1-score of 78.5% and a ROC-AUC of 0.918**, detecting 53 of 68 failures with 14 false alarms. SHAP analysis and a failure-mode analysis agree that torque and tool wear are the dominant signals. The model was saved, wrapped in a reusable prediction function and connected to a simple maintenance decision framework, and the findings were presented in a Power BI dashboard. The report closes with the limitations of a synthetic dataset and a roadmap for validating the approach on real plant data.

Keywords: predictive maintenance, machine failure prediction, class imbalance, decision tree, gradient boosting, SHAP, smart manufacturing

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1. Introduction

1.1 Background

Manufacturing plants depend on a small number of critical machines. When one of them fails unexpectedly, the consequences spread quickly: production stops, work-in-progress is delayed, delivery commitments are missed, spare parts must be rushed in, and Overall Equipment Effectiveness (OEE) falls. Traditional maintenance policies handle this in one of two ways — they repair a machine after it breaks (reactive maintenance) or they service it on a fixed calendar whether or not it needs attention (preventive maintenance). The first is costly in downtime; the second wastes effort on healthy machines and still misses failures that arrive between service dates.

Predictive maintenance takes a third route. It uses the operating data a machine already produces to estimate how likely a failure is, so that inspection can be scheduled for the machines that actually need it, before they break. Machine learning is well suited to this task because failures usually emerge from combinations of conditions — high load together with an old tool, for instance — that are hard to capture with a single alarm threshold.

1.2 Problem Statement

Production and maintenance managers face sudden breakdowns, rising maintenance expense, production delays, lower productivity, higher rejection rates and dissatisfied customers. The question this project answers is: **can the routine operating readings of a milling machine be used to predict, ahead of time, whether it is about to fail — and which readings matter most?**

1.3 Objectives

- Analyse machine operating conditions and understand how failed and healthy machines differ.
- Identify the factors most strongly associated with machine failure.
- Build and compare machine-learning classifiers that predict failure, and select the most suitable one for a maintenance setting.
- Explain the chosen model so that maintenance engineers can understand and trust its output.
- Translate predictions into practical maintenance actions and a reusable prediction tool.
- Present the findings in a dashboard that supports data-driven maintenance decisions.

1.4 Scope

The project covers the machine-learning lifecycle from business understanding to a reusable prediction system: data understanding and quality checks, exploratory and bivariate analysis, feature preparation, development and tuning of six classification models, model evaluation, SHAP explainability, root-cause analysis of failure modes, a maintenance recommendation framework, model persistence, and a Power BI dashboard. It uses a public, synthetic dataset; it does not include live sensor integration or deployment on a real production line (see Chapter 13).

1.5 Why Predictive Maintenance Matters

- **Lower maintenance cost** — effort goes to machines that show risk instead of to a fixed schedule.
- **Less downtime** — problems are found before they stop production.
- **Longer equipment life** — tools and components are replaced at the right time.
- **Better planning** — spare parts, labour and production schedules can be arranged in advance.
- **Higher productivity and quality** — fewer disruptions and fewer defects caused by worn tooling.

2. Dataset and Tools

2.1 The AI4I 2020 Predictive Maintenance Dataset

The project uses the AI4I 2020 Predictive Maintenance Dataset published by S. Matzka (2020). It is a synthetic dataset, modelled on the behaviour of a real milling machine, containing 10,000 observations and 14 variables. Each row describes the operating condition of the machine at one point in time and records whether a machine failure occurred. Failures are rare — 339 of 10,000 records (3.39%) — which reflects real manufacturing environments and makes the problem highly imbalanced.

Table 1. *Variables in the dataset*

Variable	Description	Role
UDI	Unique record identifier	Identifier (not used)
Product ID	Product identification number	Identifier (not used)
Type	Product quality category: L (low), M (medium), H (high)	Predictor
Air temperature [K]	Ambient air temperature	Predictor
Process temperature [K]	Operating process temperature	Predictor
Rotational speed [rpm]	Machine rotational speed	Predictor
Torque [Nm]	Machine torque	Predictor
Tool wear [min]	Cumulative tool wear in minutes	Predictor
Machine failure	1 if the machine failed, otherwise 0	Target
TWF / HDF / PWF / OSF / RNF	Tool wear, heat dissipation, power, overstrain and random failure indicators	Used for analysis only

2.2 Failure Modes

Table 2. *Failure modes recorded in the dataset*

Code	Failure mode	Meaning
TWF	Tool Wear Failure	The tool is worn excessively and needs replacement
HDF	Heat Dissipation Failure	Heat is not removed effectively and the machine overheats
PWF	Power Failure	The machine cannot deliver the power the operation requires
OSF	Overstrain Failure	The machine is loaded beyond its design limits
RNF	Random Failure	Unexpected failure not explained by the measured variables

2.3 Tools and Environment

Analysis and modelling were done in Python in a Jupyter Notebook. Pandas and NumPy were used for data handling, Matplotlib and Seaborn for visualisation, Scikit-learn for preprocessing, modelling, tuning and evaluation, SHAP for model explanation, and Joblib to save the trained model. Dashboards were built in Power BI.

3. Methodology

The project follows a standard machine-learning workflow. Every model was trained and evaluated on the same data split, so results can be compared fairly.

1. **Understand the data** — inspect structure, types, ranges, missing values, duplicates and the target distribution.
2. **Explore** — univariate, bivariate and correlation analysis to see how each reading relates to failure.
3. **Prepare** — select predictors, encode product type, split the data 80/20 with stratification, and scale features where an algorithm needs it.
4. **Model** — train six classifiers; tune hyper-parameters with grid search and cross-validation.
5. **Evaluate** — compare models on failure-class precision, recall and F1-score, confusion matrices, ROC and precision–recall curves.
6. **Explain** — use feature importance and SHAP to show what drives the predictions.
7. **Act** — convert predictions into maintenance recommendations, save the model, and present results in a dashboard.

3.1 Evaluation Strategy for an Imbalanced Problem

With only 3.39% failures, a model that always predicts “no failure” would be 96.6% accurate while detecting nothing. Accuracy therefore cannot be the main yardstick. For the failure class the report uses:

- **Precision** — of the machines flagged as failing, how many really were (controls false alarms and unnecessary inspections).
- **Recall** — of the machines that really failed, how many were caught (controls missed failures and unplanned downtime).
- **F1-score** — the harmonic mean of precision and recall, used as the balancing measure and as the tuning objective.
- **ROC-AUC and Average Precision** — threshold-independent measures of how well the model ranks failures above normal operation.

4. Data Understanding and Quality

4.1 Structure and Quality Checks

The dataset has 10,000 rows and 14 columns. Column names were standardised for coding consistency (for example, “Air temperature [K]” became air_temperature_k). The quality checks were clean:

- **No missing values** in any column, so no imputation or row removal was needed.
- **No duplicate records.**
- Numerical variables, the categorical product type and the target were identified and separated for later steps.

4.2 Descriptive Statistics

Table 3. *Descriptive statistics of the operating variables (10,000 records)*

Variable	Mean	Std. dev.	Min	Max
Air temperature (K)	300.00	2.00	295.3	304.5
Process temperature (K)	310.01	1.48	305.7	313.8
Rotational speed (rpm)	1,538.78	179.28	1,168	2,886
Torque (Nm)	39.99	9.97	3.8	76.6
Tool wear (min)	107.95	63.65	0	253

Temperatures are stable. Rotational speed and torque vary much more, which reflects different operating conditions, and tool wear spans the whole tool life from new (0 min) to 253 min. Rotational speed has a long right tail of high values that appear as outliers.

4.3 Target Variable

Of the 10,000 records, 9,661 (96.61%) are normal operation and 339 (3.39%) are failures. The dataset is therefore highly imbalanced, which shaped the choice of evaluation metrics (Section 3.1) and the use of stratified sampling (Chapter 6). Product types are also unevenly represented: L 60.0%, M 30.0% and H 10.0%.

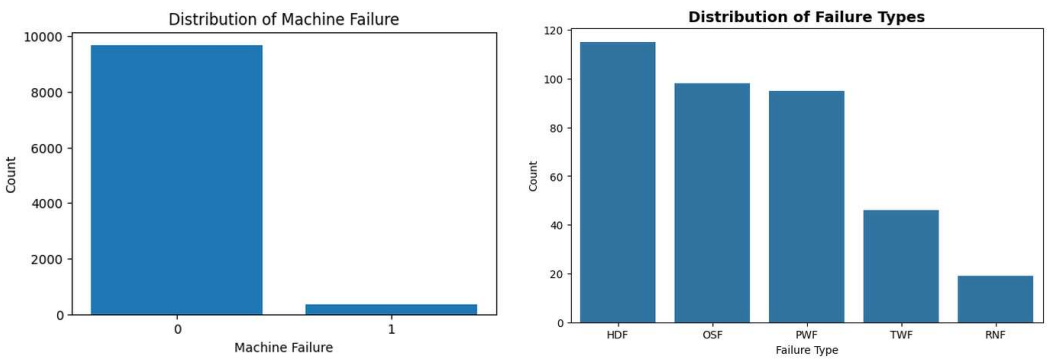


Figure 1. *Left: distribution of the target variable. Right: distribution of the five failure types.*

4.4 Correlation Analysis

The correlation matrix reveals two strong relationships between predictors: **air and process temperature** are strongly positively correlated (0.88), and **rotational speed and torque** are strongly negatively correlated (−0.88) — as speed rises, torque falls, which is consistent with a machine operating at roughly constant power. The failure-mode flags HDF (0.58), OSF (0.53) and PWF (0.52) correlate moderately with machine failure, which is expected because they define it. Tool wear correlates weakly with failure in linear terms (0.11), but this understates its importance, as later chapters show. Most remaining relationships are weak, so linear correlation alone is not enough to describe failure behaviour.

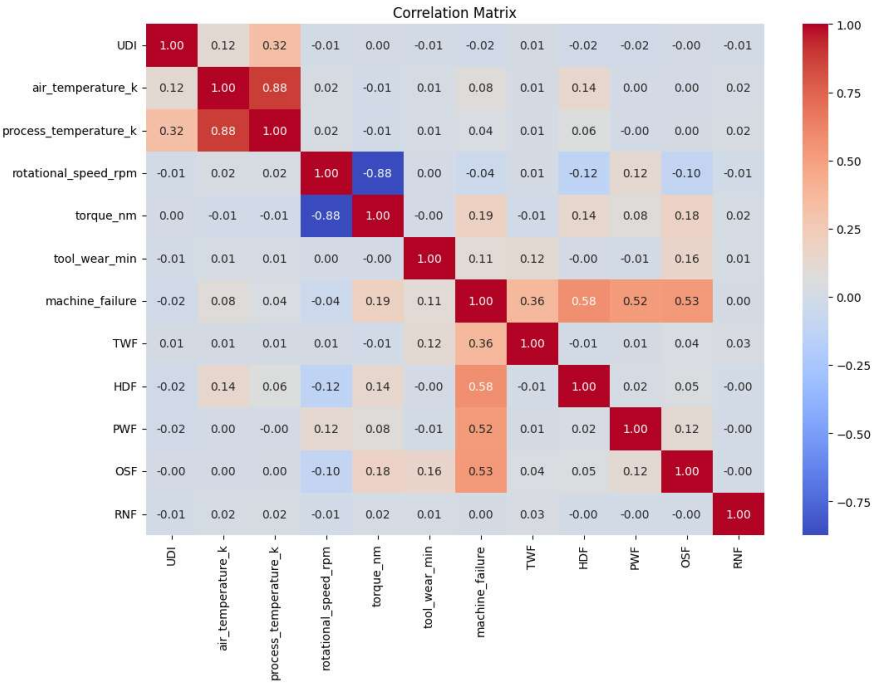


Figure 2. *Correlation matrix of the numerical variables and failure indicators.*

5. Exploratory Data Analysis

5.1 Distribution of Operating Variables

- **Air and process temperature** are roughly bell-shaped and narrow, with ranges of about 9 K and 8 K.
- **Rotational speed** is right-skewed, concentrated around 1,500 rpm, with a large number of high-speed outliers up to 2,886 rpm.
- **Torque** is close to normally distributed around 40 Nm, with outliers at both the low and high ends.
- **Tool wear** is spread almost evenly between 0 and about 200 min and then tails off, indicating that tools are observed at every stage of life.

5.2 Operating Conditions: Failed versus Normal Machines

Table 4. *Mean operating conditions of normal and failed machines*

Variable	Normal	Failure	Difference
Air temperature (K)	299.97	300.89	+0.91
Process temperature (K)	310.00	310.29	+0.29
Rotational speed (rpm)	1,540.26	1,496.49	−43.77
Torque (Nm)	39.63	50.17	+10.54
Tool wear (min)	106.69	143.78	+37.09

Failed machines run at about 27% higher torque and 35% higher tool wear than healthy ones, and slightly lower speed (about 3% lower). Temperature differences are comparatively small. Mechanical variables therefore carry more information about failure than environmental ones.

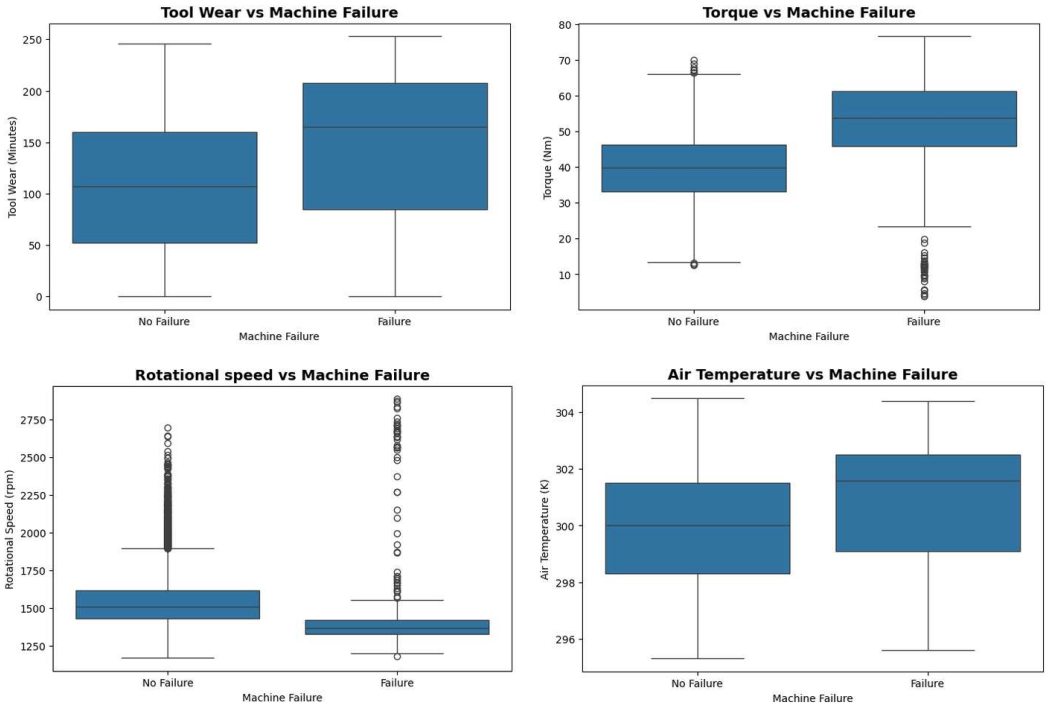


Figure 3. *Tool wear, torque, rotational speed and air temperature for normal (0) and failed (1) machines.*

5.3 Product Type

Table 5. *Failure rate by product type*

Product type	Normal	Failure	Failure rate
L (low)	5,765	235	3.92%
M (medium)	2,914	83	2.77%
H (high)	982	21	2.09%

L-type products have the highest failure rate. This may relate to their high production volume and machine utilisation, although the dataset cannot confirm the reason. Product type turns out to be the weakest predictor in the final model.

5.4 Failure Modes

Failure modes can occur together, so their counts add up to more than the 339 failed records. Heat dissipation is the most common mode, followed by overstrain and power failure; tool wear failure is comparatively uncommon and random failure is the rarest.

Table 6. *Frequency of failure modes*

Failure mode	Count	Share of failures
HDF – Heat Dissipation	115	33.9%
OSF – Overstrain	98	28.9%
PWF – Power	95	28.0%

Failure mode	Count	Share of failures
TWF – Tool Wear	46	13.6%
RNF – Random	19	5.6%

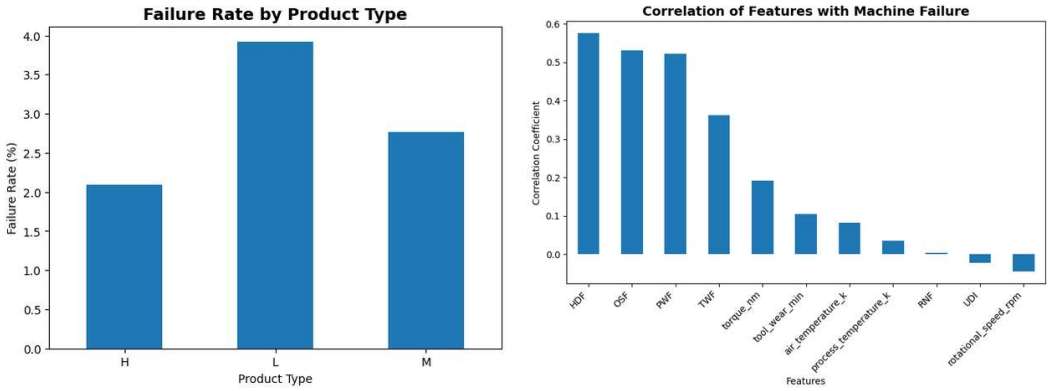


Figure 4. *Left: failure rate by product type. Right: correlation of each variable with machine failure.*

5.5 Key Insights from the Analysis

- Tool wear and torque are the strongest operating differences between failed and healthy machines.
- Lower rotational speed accompanies failure on average, though one mode (power failure) occurs at high speed.
- Temperatures have limited individual predictive power.
- L-type products show the highest failure rate.
- Mechanical load, tool condition and speed should be the focus of any monitoring strategy.

6. Data Preparation

Data was prepared once and reused for every model so that comparisons are fair.

6.1 Feature Selection

Six predictors were selected: product type, air temperature, process temperature, rotational speed, torque and tool wear. The target is machine failure. Identifier columns were dropped. The five failure-mode flags were not used as predictors, because they define the target itself and would not be known before a failure occurs.

6.2 Encoding

Product type was converted to numbers with label encoding (H = 0, L = 1, M = 2) so that every algorithm can process it.

6.3 Train–Test Split

The data was split 80/20 with a fixed random state (42) and stratification, giving 8,000 training and 2,000 test records. Stratification preserves the 3.4% failure rate in both subsets, so the test set contains 68 failures and 1,932 normal records.

6.4 Feature Scaling

Standardisation (zero mean, unit variance) was fitted on the training data and applied to both subsets for the algorithms that depend on distances or gradients — K-Nearest Neighbours, Logistic Regression and Support Vector Machine. Tree-based models do not need scaling.

7. Model Development

Six classification algorithms were developed. For each, a baseline model was trained first and then tuned with GridSearchCV (cross-validation, F1-score as the objective). All figures below are for the failure class on the 2,000-record test set unless stated otherwise.

7.1 Logistic Regression

Logistic Regression is a linear baseline. The baseline model reached 96.80% accuracy but detected only 8 of 68 failures (recall 0.12). Grid search selected C = 10, L1 penalty and the liblinear solver, which changed little (recall 0.13, F1 0.22). Setting class_weight to “balanced” was far more influential: recall rose to 0.84 (57 of 68 failures found) but accuracy fell to 82.0% because of many false alarms. This illustrates the central trade-off of the project — catching more failures costs more false inspections.

7.2 K-Nearest Neighbours

KNN classifies a machine by looking at its most similar neighbours in the scaled feature space. The baseline (K = 7) reached 97.25% accuracy with precision 0.81, recall 0.25 and F1 0.38. Testing values of K identified K = 5 as best; the tuned model reached 97.40% accuracy, precision 0.83, recall 0.29 and F1 0.43. KNN is precise but misses most failures, because failures are rare and their neighbourhoods are dominated by normal records.

7.3 Decision Tree

A Decision Tree learns a hierarchy of if–then rules, which makes it easy to explain to maintenance engineers. The unconstrained baseline fitted the training data perfectly (100% accuracy) and reached 97.95% on the test set with precision 0.68, recall 0.75 and F1 0.71 — a sign of over-fitting. Grid search selected the Gini criterion, **maximum depth 10, minimum samples per leaf 4** and minimum samples to split 2 (cross-validated F1 0.7194). The tuned tree trained to 98.99% and tested at **98.55%**, with precision 0.79, recall 0.78 and F1 0.79: a smaller train–test gap and better failure detection.

7.4 Random Forest

Random Forest averages 100 decision trees to reduce variance. The baseline reached 98.50% accuracy with the highest precision of all models (0.88), but recall of only 0.65 (44 of 68 failures found, 24 missed). Grid search returned the default configuration (100 trees, no depth limit, square-root feature sampling, no class weights) as best, so tuning gave no further gain.

7.5 Support Vector Machine

Linear, RBF, polynomial and sigmoid kernels were compared on scaled data. After tuning, the best model used an RBF kernel with C = 100 and gamma = “scale” (cross-validated F1 0.6901). Tuning raised recall from 0.21 to 0.63 and F1 from 0.33 to 0.69, at the cost of some precision (0.88 to 0.75); test accuracy was 98.05%.

7.6 Gradient Boosting

Gradient Boosting builds trees one after another, each correcting the errors of the previous ones. Tuning selected a learning rate of 0.1, maximum depth 4, minimum samples per leaf 2, minimum samples to split 5 and 100 estimators (cross-validated F1 0.7552). Compared with its baseline (precision 0.87, recall 0.66, F1 0.75), the tuned model improved recall to 0.72 and F1 to 0.78, with test accuracy of 98.60% — the highest of all models. It detected 49 of 68 failures with 9 false alarms.

8. Model Comparison and Selection

8.1 Overall Comparison

Table 7. *Test-set performance of all models (failure class)*

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression*	96.80%	0.67	0.12	0.20
KNN*	97.25%	0.81	0.25	0.38
Tuned Decision Tree	98.55%	0.79	0.78	0.79
Random Forest	98.50%	0.88	0.65	0.75
Tuned SVM	98.05%	0.75	0.63	0.69
Tuned Gradient Boosting	98.60%	0.84	0.72	0.78

** Baseline results as recorded in the comparison. Tuned values: Logistic Regression (C = 10, L1) accuracy 96.80%, precision 0.64, recall 0.13, F1 0.22; KNN (K = 5) accuracy 97.40%, precision 0.83, recall 0.29, F1 0.43. Neither changes the ranking.*

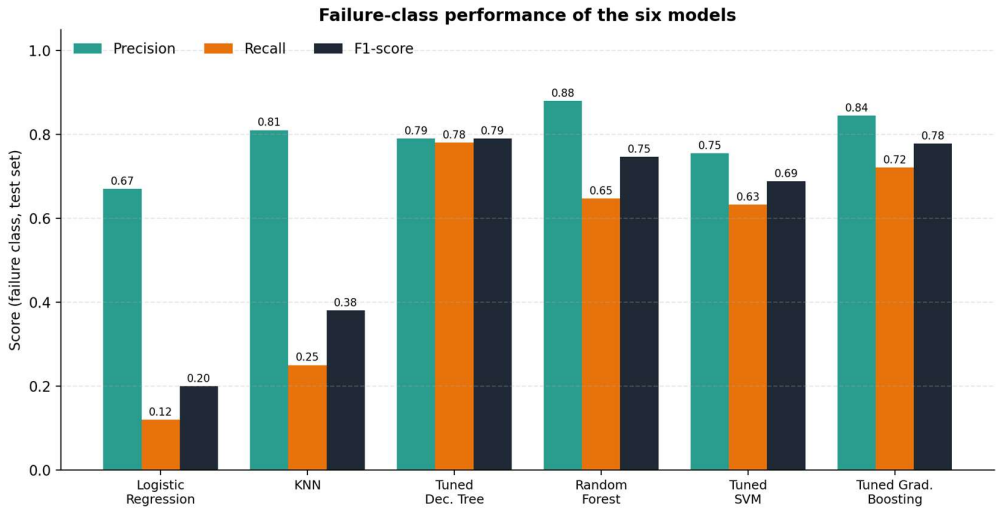


Figure 5. *Failure-class precision, recall and F1-score of the six models.*

Every model is above 96% accurate, which confirms that accuracy cannot separate them. The differences appear on the failure class:

- **Tuned Decision Tree** has the highest recall (0.78) and F1-score (0.79).
- **Tuned Gradient Boosting** has the highest accuracy (98.60%) and an F1-score only about 1.2 points lower (0.78).
- **Random Forest** has the highest precision (0.88) — when it raises an alarm it is usually right — but misses more failures.
- **Logistic Regression and KNN** are not suitable in their default form: they miss three-quarters or more of all failures.

8.2 Final Model Selection

The **tuned Decision Tree** was selected as the final model for three reasons: it offers the best balance between catching failures and avoiding false alarms (highest F1 and recall); it is transparent, so its rules and decision paths can be shown to engineers; and it generalises well, with training accuracy of 98.99% against test accuracy of 98.55%. Gradient Boosting remains a strong alternative if the highest overall accuracy is preferred.

Interpretation note: the test set contains only 68 failures, so a difference of four failures caught (Decision Tree 53, Gradient Boosting 49) is small relative to sampling noise. The ranking between the top models should be confirmed with cross-validation or additional data before deployment.

8.3 Final Evaluation of the Selected Model

Table 8. *Tuned Decision Tree – confusion matrix and metrics on the test set*

Measure	Value
Correctly identified normal records	1,918
Correctly detected failures (true positives)	53
False alarms (false positives)	14
Missed failures (false negatives)	15
Accuracy	98.55%
Precision (failure)	79.1%
Recall (failure)	77.9%
F1-score (failure)	78.5%
ROC-AUC	0.9184
Average precision (baseline prevalence 0.034)	0.7536

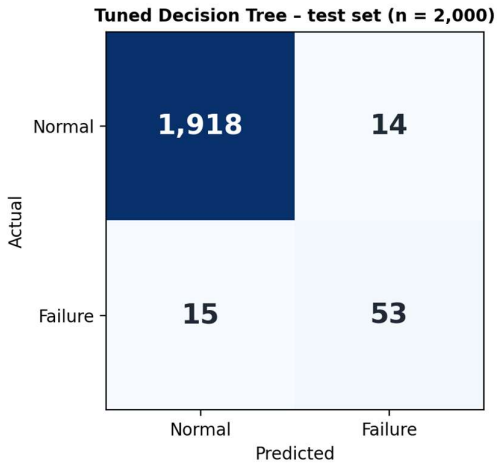


Figure 6. *Confusion matrix of the tuned Decision Tree on the 2,000-record test set.*

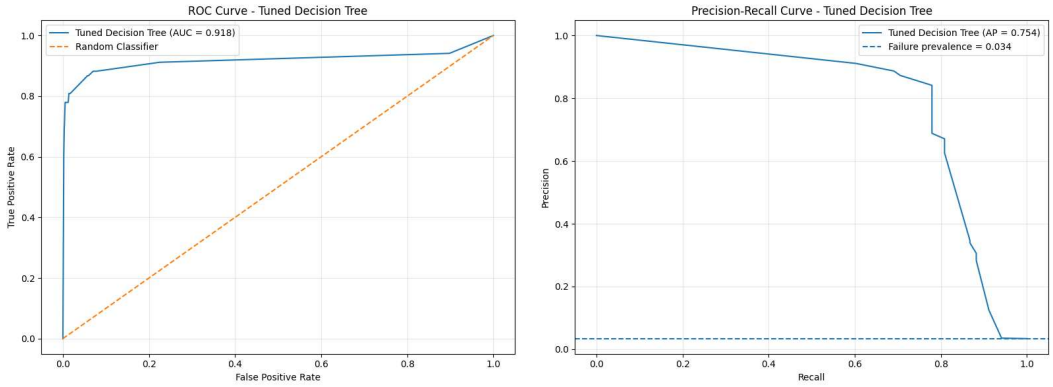


Figure 7. *ROC curve (AUC = 0.918) and precision–recall curve (AP = 0.754) of the tuned Decision Tree.*

The ROC-AUC of 0.918 shows strong ability to rank failing machines above normal ones. The average precision of 0.754 is more than twenty times the 0.034 failure prevalence, which is the relevant benchmark for an imbalanced problem. In operational terms, the model would have flagged 67 machines for inspection, of which 53 were genuine failures, while 15 genuine failures were missed. Those 15 misses are why the model should be treated as decision support rather than a replacement for engineering inspection.

9. Model Explainability

A maintenance team will only act on a model it can understand. Two complementary techniques were used.

9.1 Decision Tree Feature Importance

Table 9. *Feature importance in the baseline Decision Tree*

Rank	Feature	Importance
1	Torque	0.345
2	Air temperature	0.207
3	Tool wear	0.143
4	Rotational speed	0.140
5	Process temperature	0.127
6	Product type	0.038

Torque is the most influential feature. The unexpectedly high rank of air temperature should be read with care: it comes from the unpruned baseline tree, which over-fits, and it is not supported by the exploratory analysis, where temperature differences between failed and healthy machines are small.

9.2 SHAP Analysis

SHAP (SHapley Additive exPlanations) was applied to the tuned Decision Tree. It attributes each prediction to the contribution of every input, giving both a global ranking and an explanation for individual machines.

Table 10. *Global SHAP ranking (mean absolute SHAP value)*

Rank	Feature	Mean |SHAP|
1	Torque	0.8574
2	Tool wear	0.0716
3	Rotational speed	0.0202
4	Air temperature	0.0188
5	Process temperature	0.0028
6	Product type	0.0011

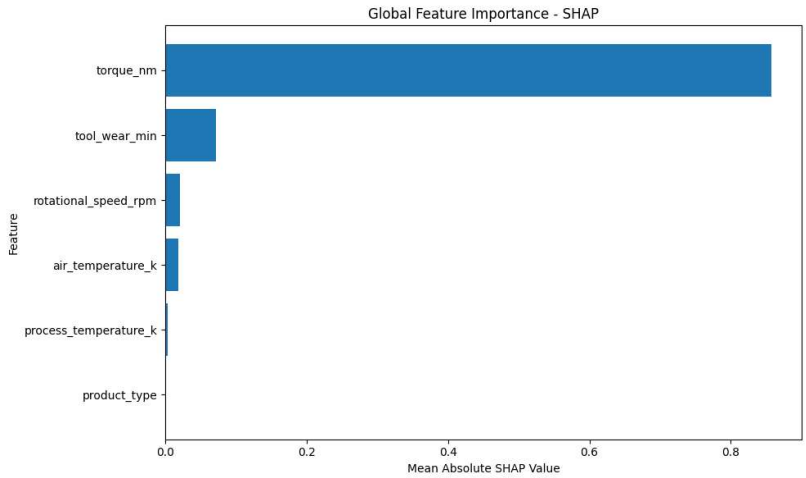


Figure 8. *Global feature importance from SHAP for the tuned Decision Tree.*

Torque dominates, followed by tool wear. Rotational speed has a smaller but noticeable effect, temperatures a minor one, and product type almost none. A local explanation was produced for machine 1391, a correctly predicted failure: torque contributed +0.857 to the failure prediction and tool wear +0.071, with the other features adding little.

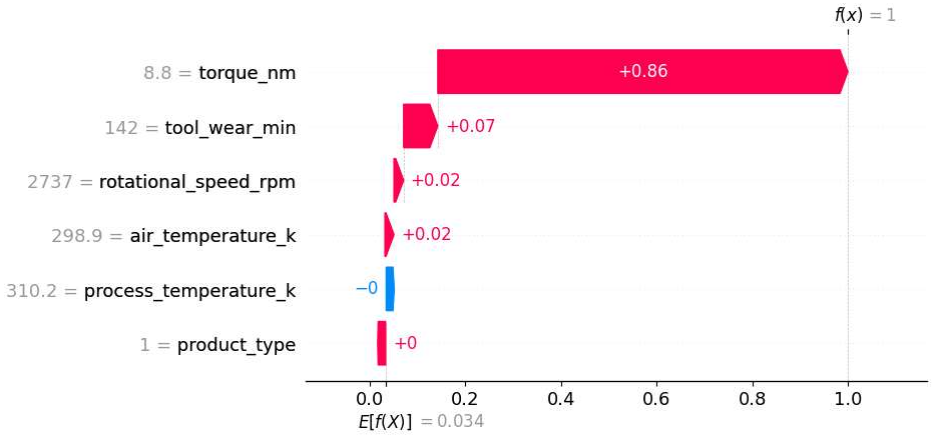


Figure 9. *SHAP waterfall plot for machine 1391 (actual failure, predicted failure).*

SHAP explains the patterns the model has learned. It shows which readings drive predictions, but it does not prove that those readings physically cause failure.

10. Root-Cause and Failure-Risk Analysis

To connect predictions with engineering action, failures were examined by mode. The average operating conditions of each failure mode reveal distinct signatures.

Table 11. *Average operating conditions by failure mode*

Mode	Air temp (K)	Speed (rpm)	Torque (Nm)	Tool wear (min)	Key pattern
TWF	300.30	1,566	37.8	216.4	Very high tool wear
HDF	302.56	1,337	53.2	107.2	High torque, low speed, warmer air
PWF	300.08	1,764	48.5	101.9	Highest rotational speed
OSF	300.04	1,350	58.4	207.7	Highest torque with high tool wear
RNF	300.82	1,485	43.7	124.5	Moderate conditions; no clear signature

This explains why single thresholds are inadequate: the failure modes occur under very different conditions — worn tools, overload, over-speed or overheating — and a useful system must watch several readings together. It also supports tailoring the maintenance response to the expected failure mode instead of applying one schedule to every machine.

Table 12. *Monitoring focus and maintenance response by failure mode*

Mode	Primary monitoring factor	Maintenance focus
TWF	Tool wear	Inspect or replace the tool
HDF	Temperature / torque	Check cooling and thermal conditions
PWF	Rotational speed	Check speed and power conditions
OSF	Torque / tool wear	Inspect the tool and mechanical load
RNF	Multiple operating conditions	Perform a broader machine inspection

These are associations found in the data and are not proof of direct causation.

11. Maintenance Recommendations and Prediction System

11.1 Recommendations by Operating Parameter

Table 13. *Maintenance recommendations by monitored parameter*

Parameter	Observed pattern	Maintenance action
Torque	Higher in failed machines	Monitor abnormal load conditions
Tool wear	Much higher in failed machines	Schedule tool inspection or replacement
Rotational speed	Lower overall in failed machines; very high for power failures	Monitor abnormal speed conditions
Air temperature	Slightly higher in failed machines	Monitor thermal conditions
Process temperature	Slightly higher in failed machines	Check cooling and process conditions

11.2 Risk Classification

Model output was converted into a two-level risk classification. Applied to the 2,000 test machines with the tuned Gradient Boosting model, 1,942 were classified **Normal** (continue routine monitoring) and 58 **Failure Risk** (prioritise inspection and preventive maintenance). A prediction is a risk signal, not a confirmed failure; the final decision should combine it with engineering inspection and machine history.

11.3 Saved Model and Reusable Prediction Function

The selected Decision Tree was saved with Joblib and reloaded to confirm that it reproduces the original predictions and expects the same six features in the same order. A reusable function takes a machine's product type, air temperature, process temperature, rotational speed, torque and tool wear, and returns the predicted status, the failure probability and a readable message. A second function adds the maintenance action: "Priority inspection recommended" when failure is predicted, or "Continue routine monitoring" otherwise.

11.4 Scenario Testing

Three hypothetical operating scenarios were used to test the system. Estimated failure probability rose from 0.04% (low torque, low wear) to 0.34% (higher torque and wear) and 50.0% under the most severe condition (torque 60 Nm, tool wear 220 min, 1,800 rpm). The pattern behaves as engineering intuition suggests, although Decision Tree probabilities come from the training samples in each leaf and are not calibrated real-world failure probabilities.

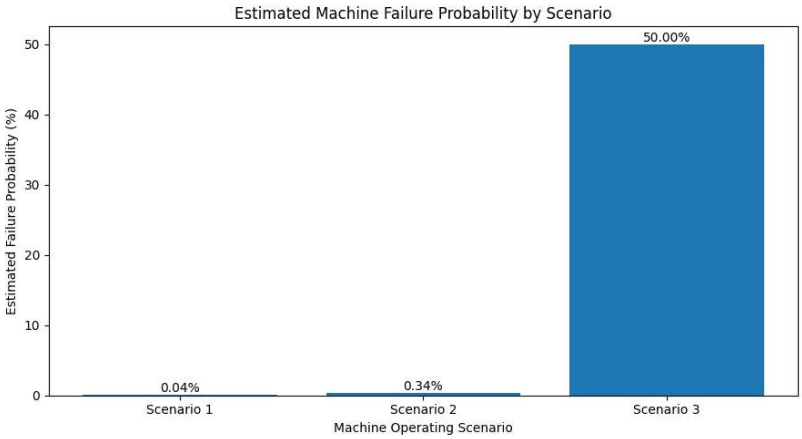


Figure 10. *Estimated failure probability for three hypothetical operating scenarios.*

12. Business Intelligence Dashboard

Results were presented in a three-page Power BI dashboard to make them accessible to non-technical users such as production and maintenance managers.

- **Machine health and failure analysis** — total records, normal and failed counts, the 3.39% failure rate, failures by product type and failure-mode indicators.
- **Operating parameter analysis** — average torque, tool wear, rotational speed and temperatures for normal and failed machines.
- **Model insights and maintenance action plan** — model comparison, feature importance and the recommended maintenance actions.

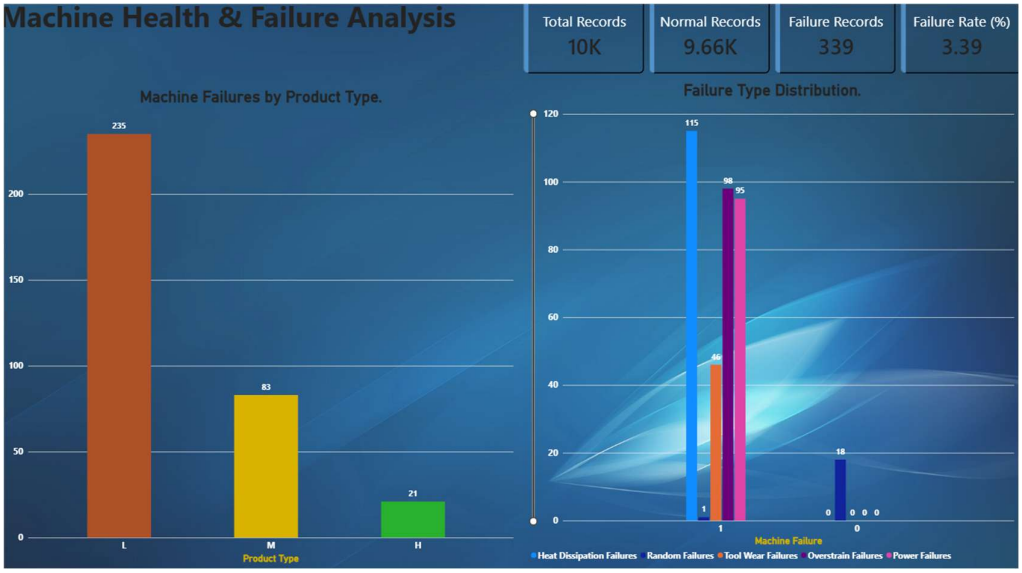


Figure 11. *Power BI dashboard – machine health and failure analysis.*

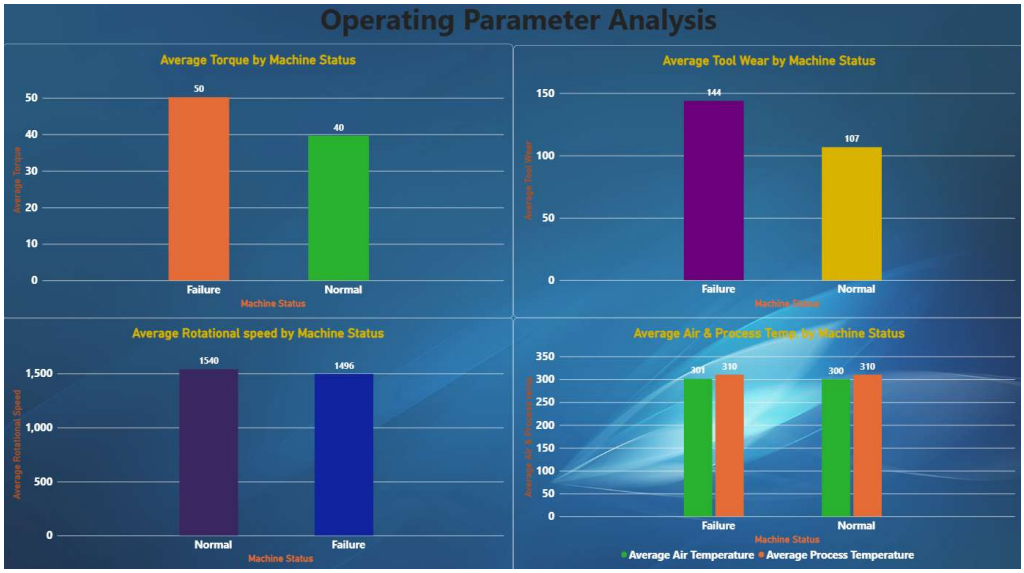


Figure 12. *Power BI dashboard – operating parameter analysis.*

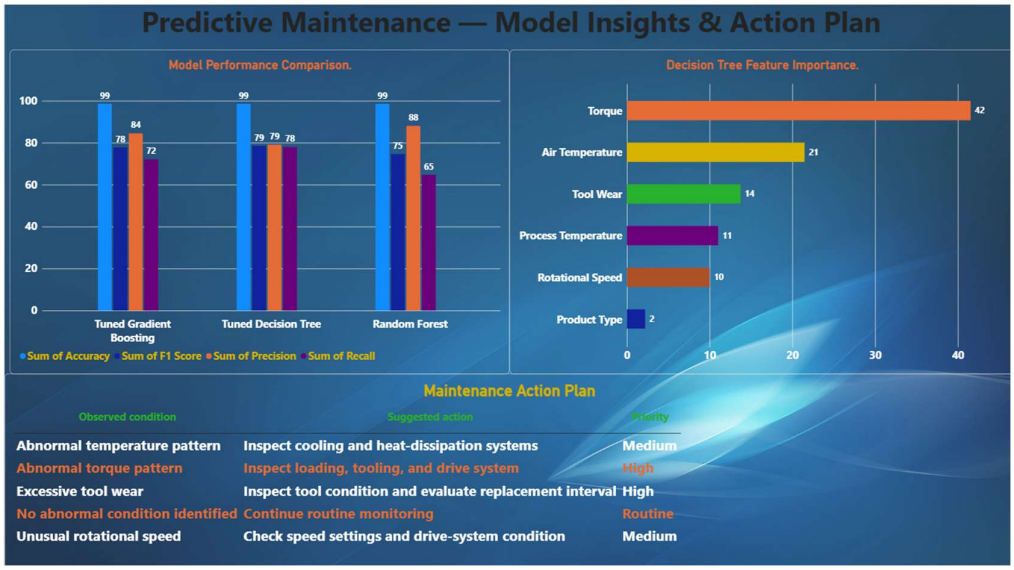


Figure 13. *Power BI dashboard – model insights and maintenance action plan.*

Failure-mode indicators can overlap, so their individual counts should not be read as mutually exclusive categories.

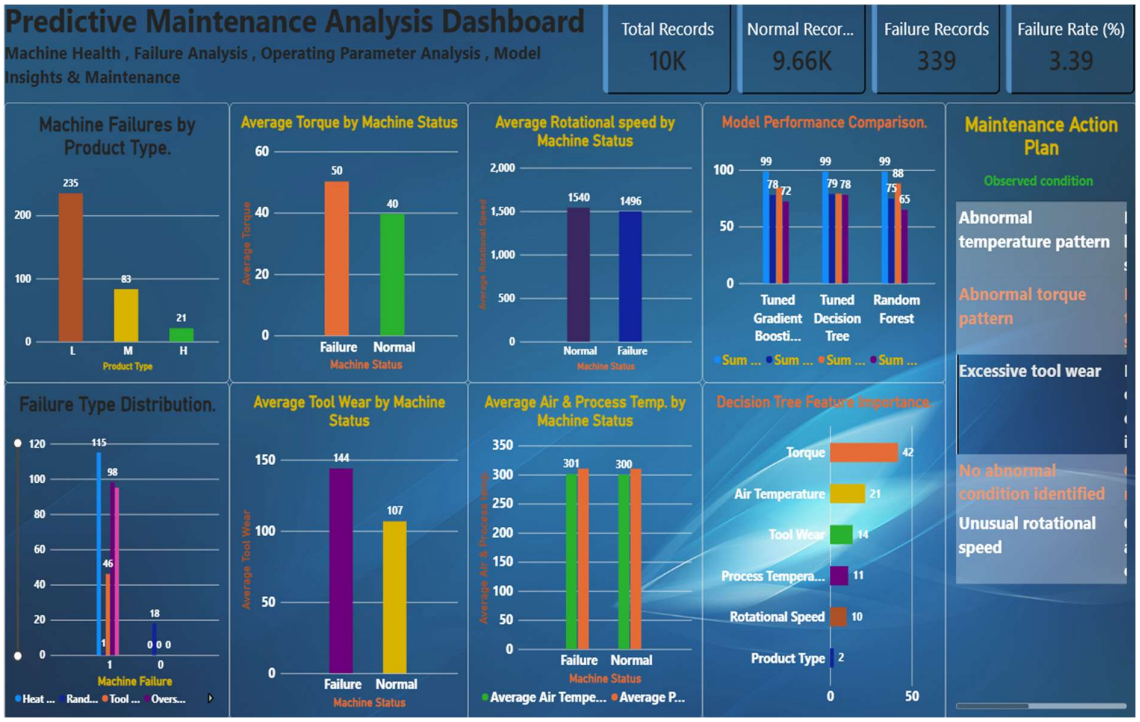


Figure 14. *Power BI dashboard – Complete Predictive Maintenance Analysis Dashboard*

13. Limitations and Future Scope

13.1 Limitations

- **Class imbalance.** With 339 failures, the test set holds only 68, so metrics are sensitive to a few cases and accuracy alone is misleading.
- **Synthetic data.** The dataset is modelled on a real milling machine but is not real plant data; performance on actual equipment may differ.
- **Probability calibration.** Decision Tree probabilities should not be read as calibrated failure probabilities.
- **Limited sensors.** Six readings are used; vibration, acoustic signals, lubrication condition and maintenance history could improve prediction.
- **Prediction is not diagnosis.** A predicted failure reflects a learned pattern; it does not confirm the physical root cause.
- **Missed failures.** The selected model still misses 15 of 68 failures, so it must support, not replace, inspection procedures.

13.2 Future Scope

8. Collect and validate real-world machine and maintenance data.
9. Apply class weighting or resampling, and tune the decision threshold, to raise failure recall at an acceptable false-alarm rate.
10. Evaluate probability calibration.
11. Add vibration, temperature-trend and maintenance-history features.
12. Use time-based validation once time-ordered production data is available.
13. Build an automated monitoring service that scores new sensor readings and integrates with maintenance workflows.
14. Monitor the model after deployment and retrain as operating conditions change.

14. Conclusion

This project built an end-to-end predictive maintenance workflow on the AI4I 2020 dataset, covering data validation, exploratory analysis, six tuned machine-learning models, explainability, root-cause analysis, a maintenance decision framework, a reusable prediction function and a Power BI dashboard.

The main findings are:

- Failures are rare (3.39%), so models must be judged on failure-class recall, precision and F1 rather than accuracy.
- Torque and tool wear are the strongest signals of failure. Failed machines averaged about 50 Nm and 144 min of tool wear, against 40 Nm and 107 min for healthy machines; temperature mattered much less.
- The tuned Decision Tree gave the best balance of performance and interpretability: 98.55% accuracy, 79.1% precision, 77.9% recall, F1 of 78.5% and ROC-AUC of 0.918.
- Different failure modes show different operating signatures, so maintenance should be targeted by mode rather than scheduled uniformly.

With validation on real plant data and integration into maintenance workflows, this approach can help teams identify risky operating conditions, prioritise inspections, reduce unplanned downtime and improve equipment availability. The model should be used as a decision-support tool alongside engineering judgement, and its benefits should be confirmed through real-world trials against measurable maintenance indicators.

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